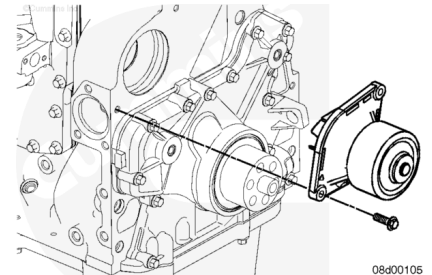


Trainings ▾

General Information

The water pump is a belt-driven, centrifugal-type pump with the inlet and bypass as integral parts of the cylinder block.

Note : It is **not** practical to replace the components of the water pump; the water pump is serviced as an assembly.

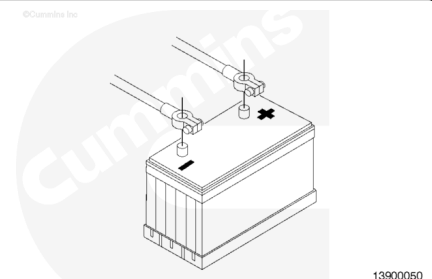


Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to the OEM service manual.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>)
- Remove the drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)

Initial Check

Inspect the water pump housing for cracks and/or damage.

Check the water pump seal weep hole. The water pump seal design requires a coolant film for lubrication and cooling. Therefore, it is normal to observe a minor chemical buildup or streaking at the weep hole.

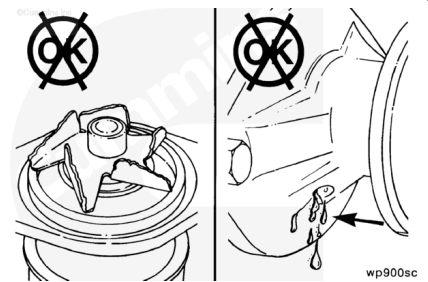
Use the following guidelines to determine if water pump replacement is necessary:

Note : A streak or chemical buildup at the weep hole is not justification for water pump replacement.

Make sure the weep hole is open.

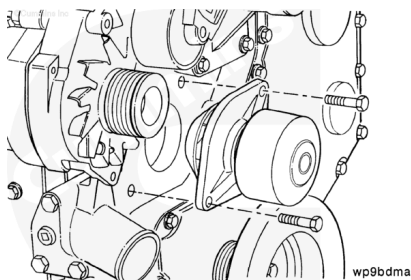
Note : A small screwdriver or a similar tool can be used to remove any debris.

If no leakage is observed from the weep hole under operating conditions, do **not** replace the water pump.



Remove

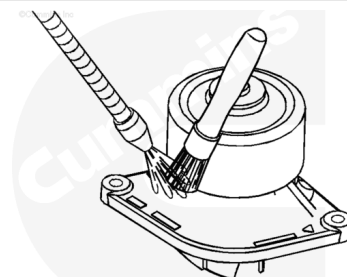
Remove the two mounting capscrews, water pump, and seal.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

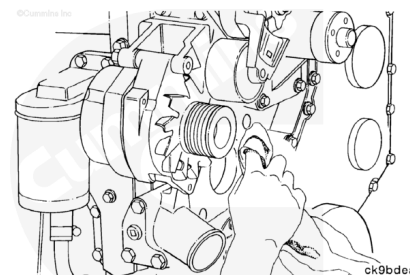


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

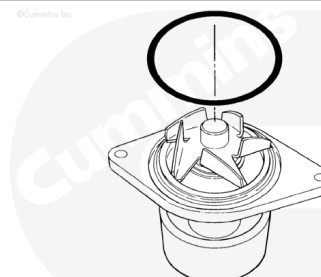
Clean the water pump with solvent. Dry with compressed air.

Clean the sealing surface on the cylinder block.



Clean the o-ring sealing surface on the water pump housing.

Inspect the sealing surface for damage. Replace the water pump if any damage is found.

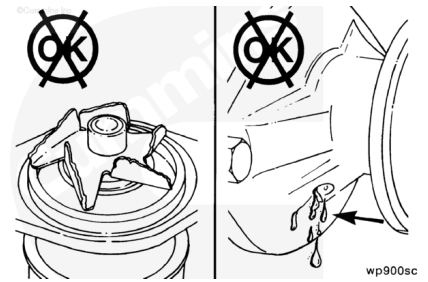


Inspect the impeller for cracks, missing blades, slippage on the shaft, and other types of damage.

Note : If any damage to the impeller blades is found, make sure to inspect the cylinder block for damage.

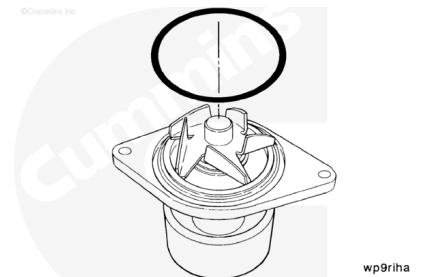
Inspect the water pump housing for damage and cracks.

Replace the water pump if any damage is found.



Install

Install the new sealing ring into the pump groove.



Install the water pump (with seal) and mounting capscrews.

Torque Value:

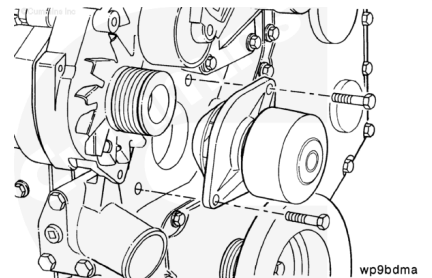
9.8 Grade

1. 24 n•m [212 in-lb]

Torque Value:

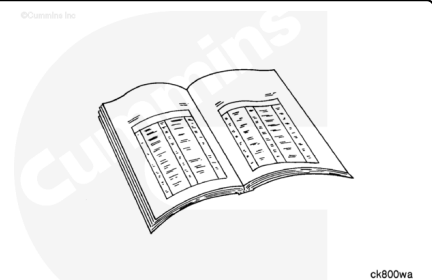
10.9 Grade

1. 30 n•m [22 ft-lb]



Finishing Steps

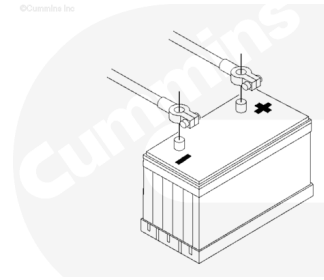
- Install the drive belt. Refer to Procedure Procedure 008-002 in Section 8.
(/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html)
- Fill the cooling system. Refer to Procedure Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html)



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for leaks.



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